

PySigns

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A CNN AI MODEL DESIPHERING ASL ALPHABETS



Agenda

03

Project Overview

04

**Model
Architecture**

11

**Dataset
Processing and
Analysis**

14

Demo

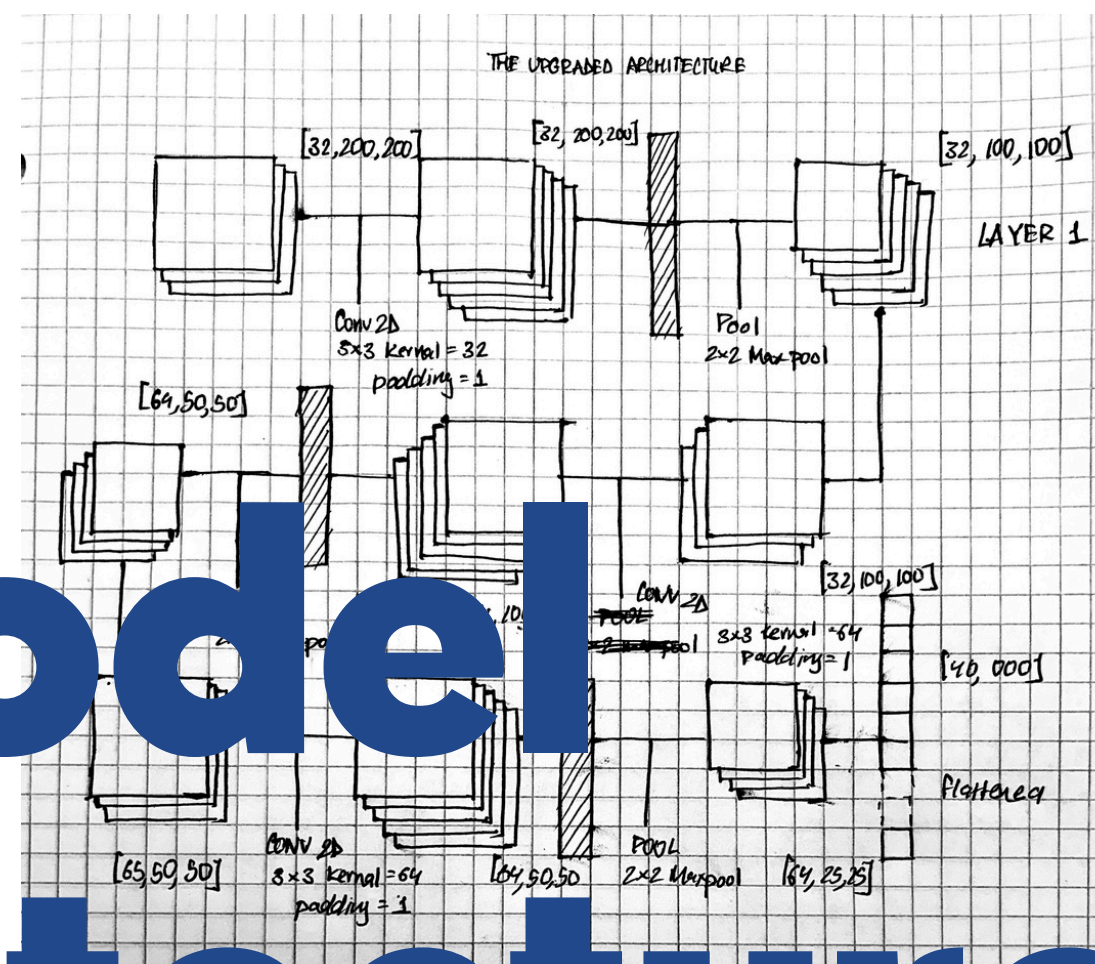
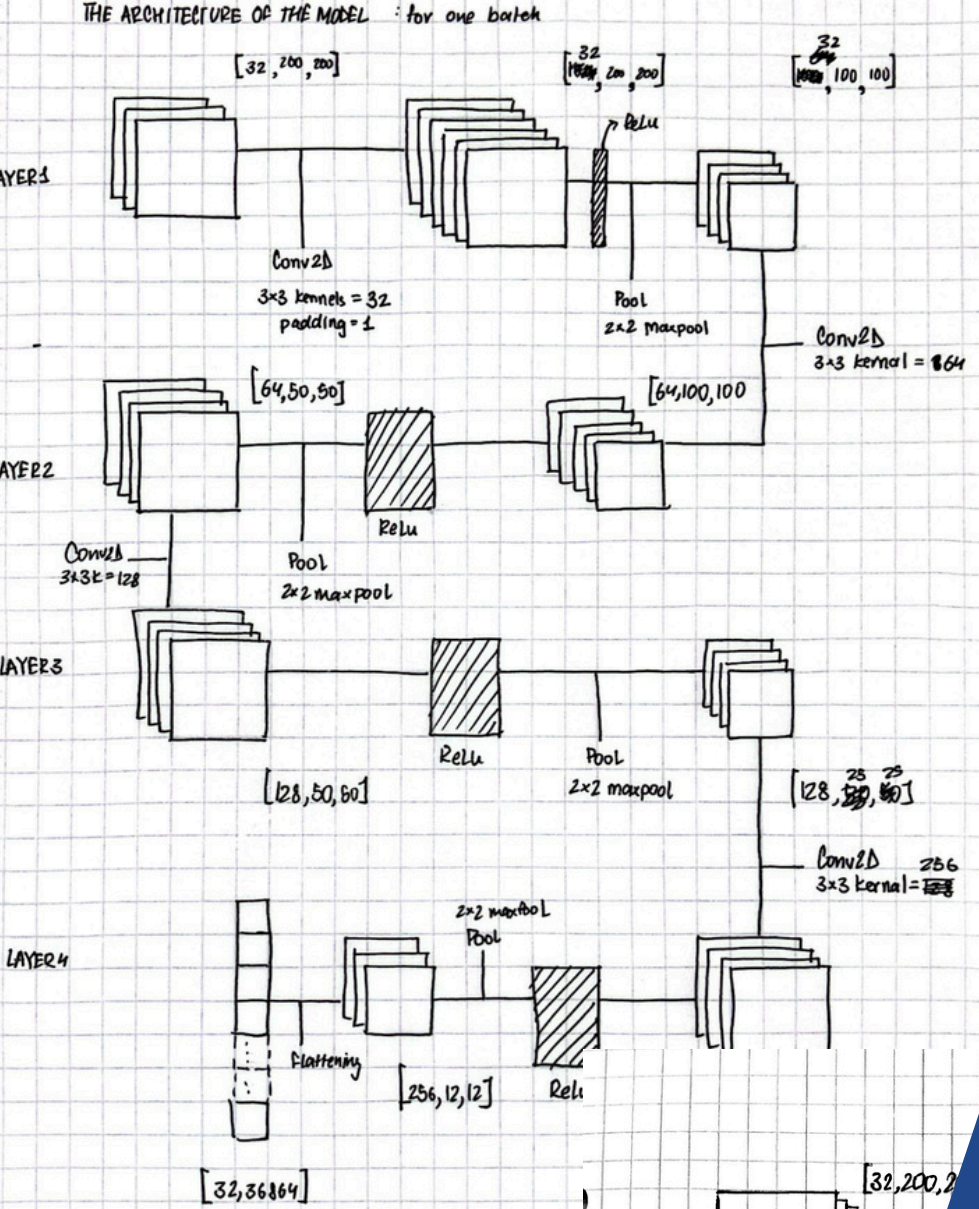
16

**Challenges/
Reflection**

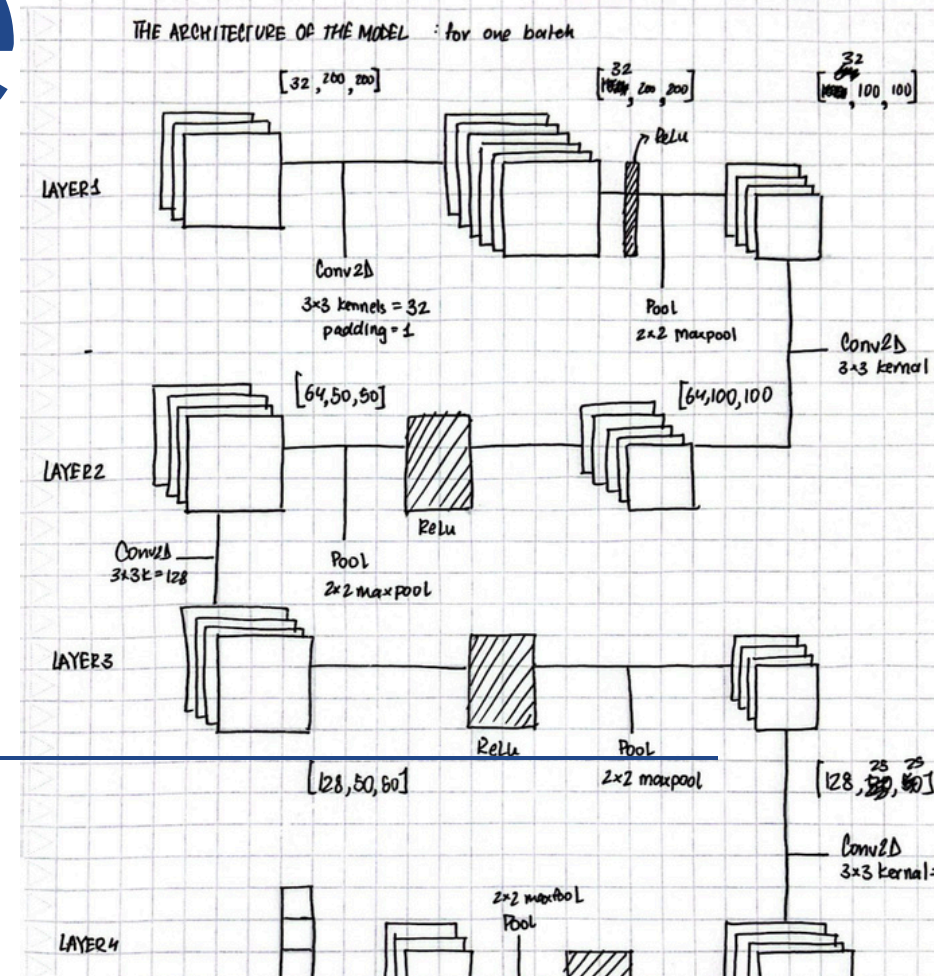
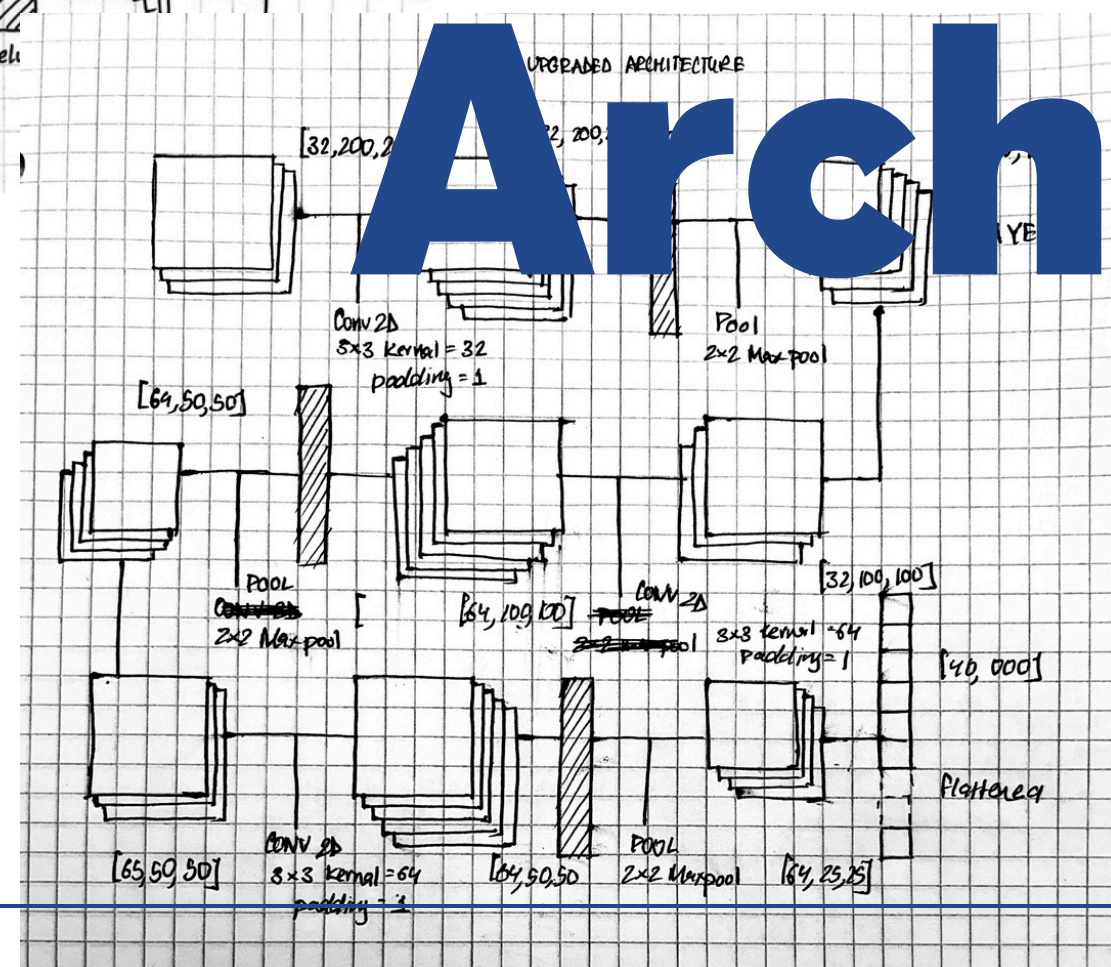


What is PySigns?

Translating/Decoding American Sign Language (ASL)
from images or hand signs
to the alphabet



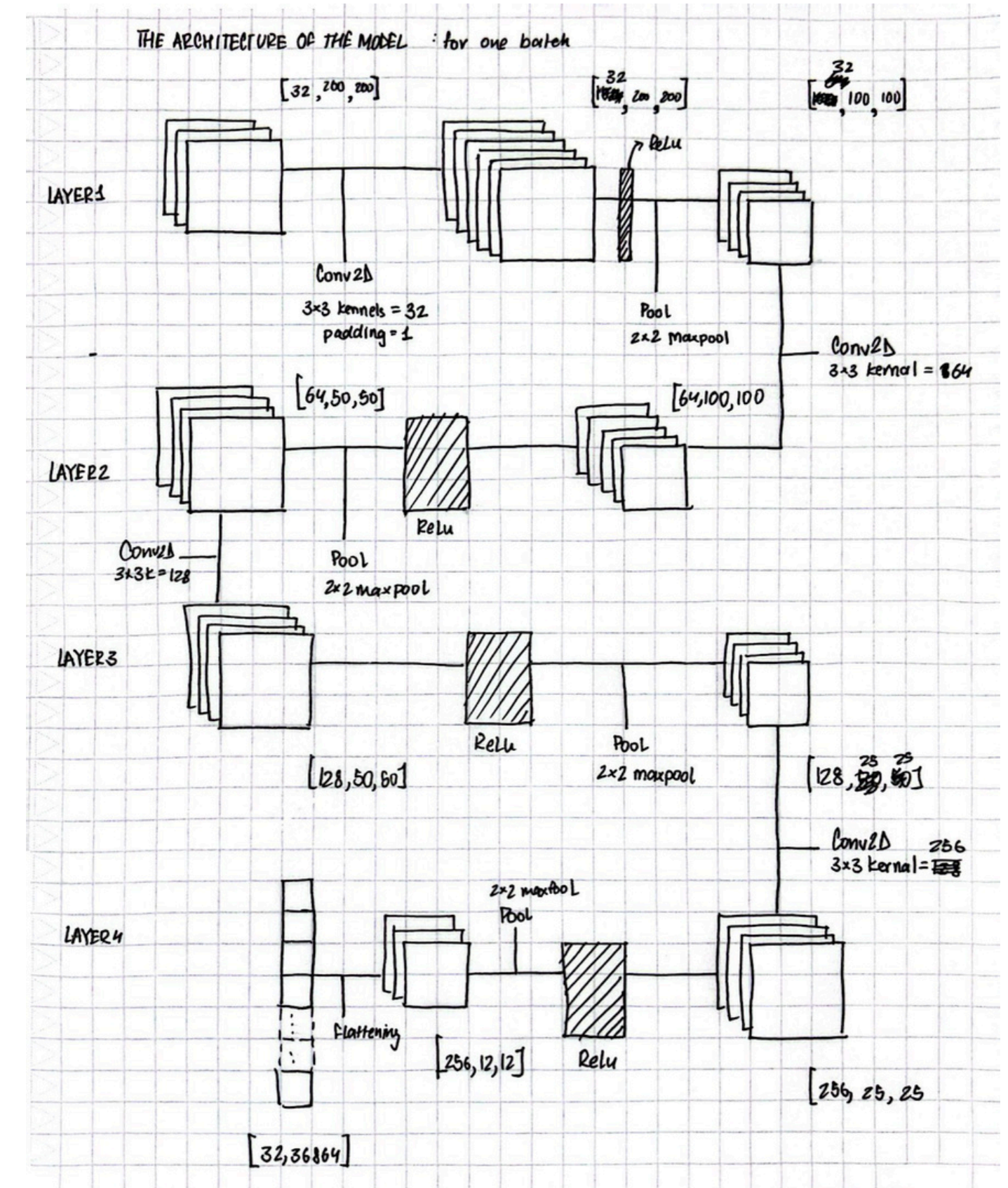
Model Architecture



Initial method

The initial model we designed had the following features :

- 1st Layer: Convolution with 32 3x3 kernels
- 2nd Layer: Convolution with 64 3x3 kernels
- 3rd Layer: Convolution with 128 3x3 kernels
- 4th Layer: Convolution with 256 3x3 kernels

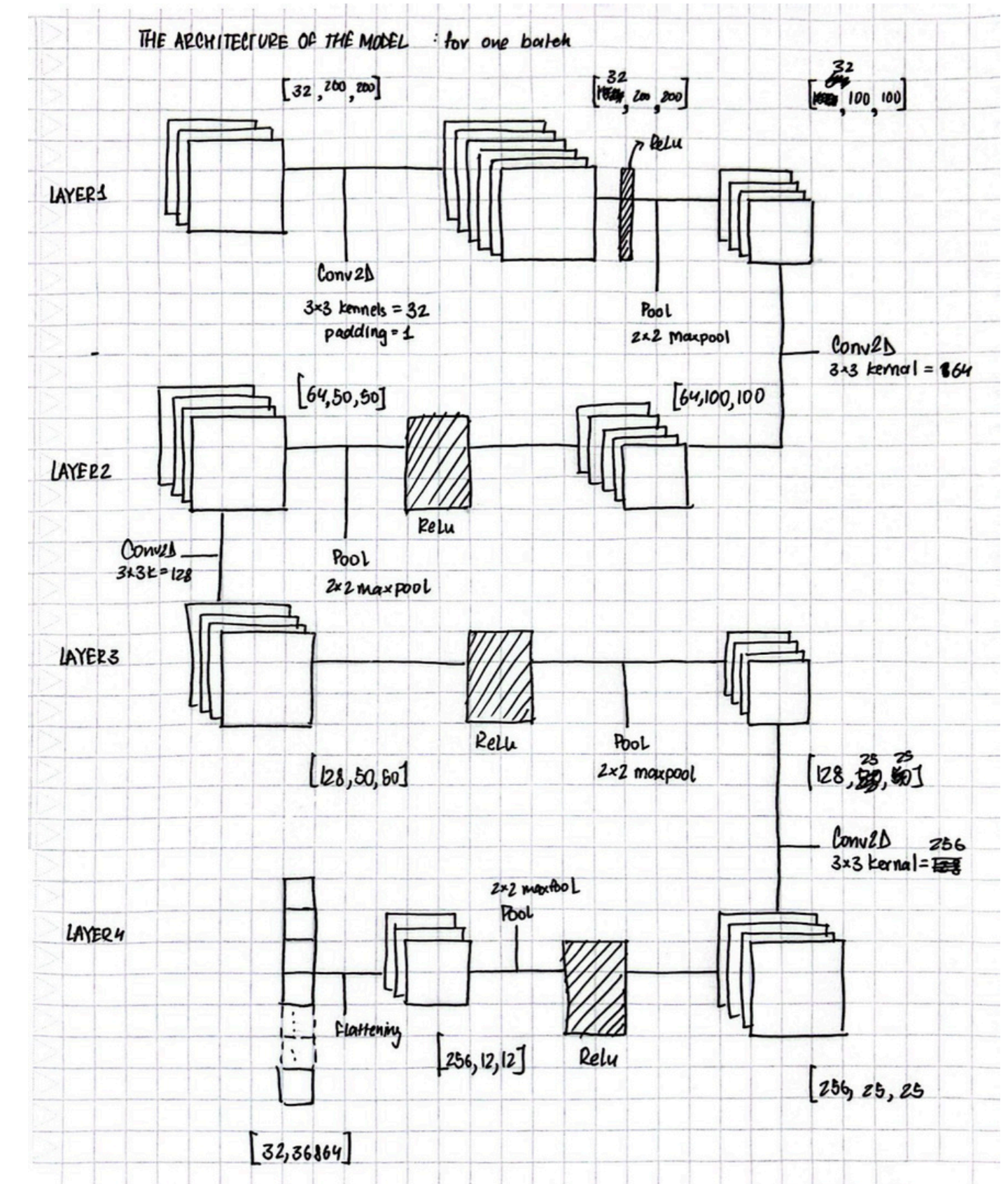


Initial method

The initial model we designed had the following features :

- 1st Layer: Convolution with 32 3x3 kernels
- 2nd Layer: Convolution with 64 3x3 kernels
- 3rd Layer: Convolution with 128 3x3 kernels
- 4th Layer: Convolution with 256 3x3 kernels

Which lead to:



The background is a solid dark blue. It is decorated with numerous small, colorful confetti pieces in shades of yellow, red, green, and blue. There are also several thin, curved streamers in blue, yellow, and red scattered across the scene. The main text is centered and reads "Overfitting!" in a large, white, sans-serif font. The exclamation mark is a solid yellow circle.

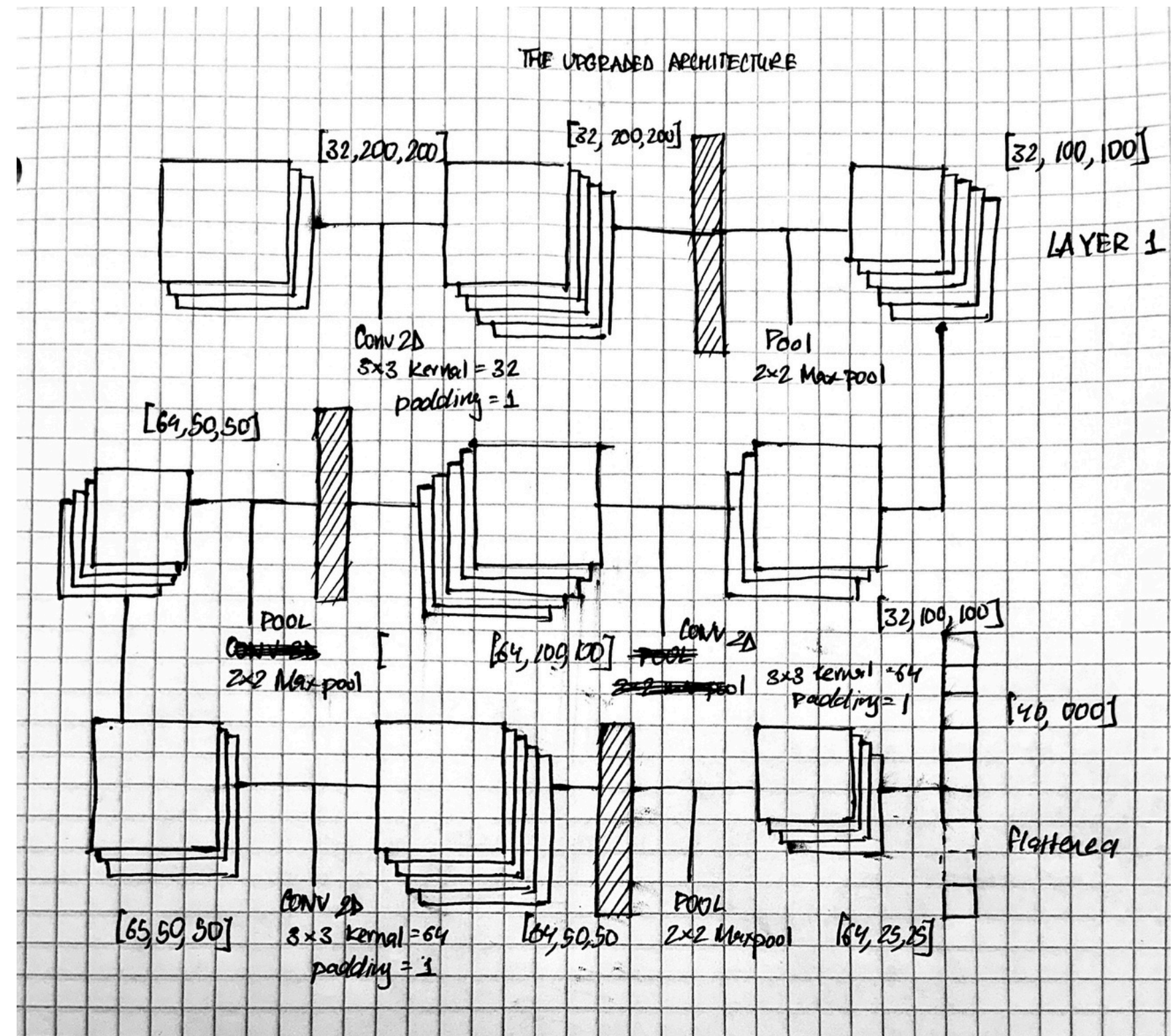
Overfitting!

With an accuracy of ~~99.37%~~

New Method

The final model we designed had the following features :

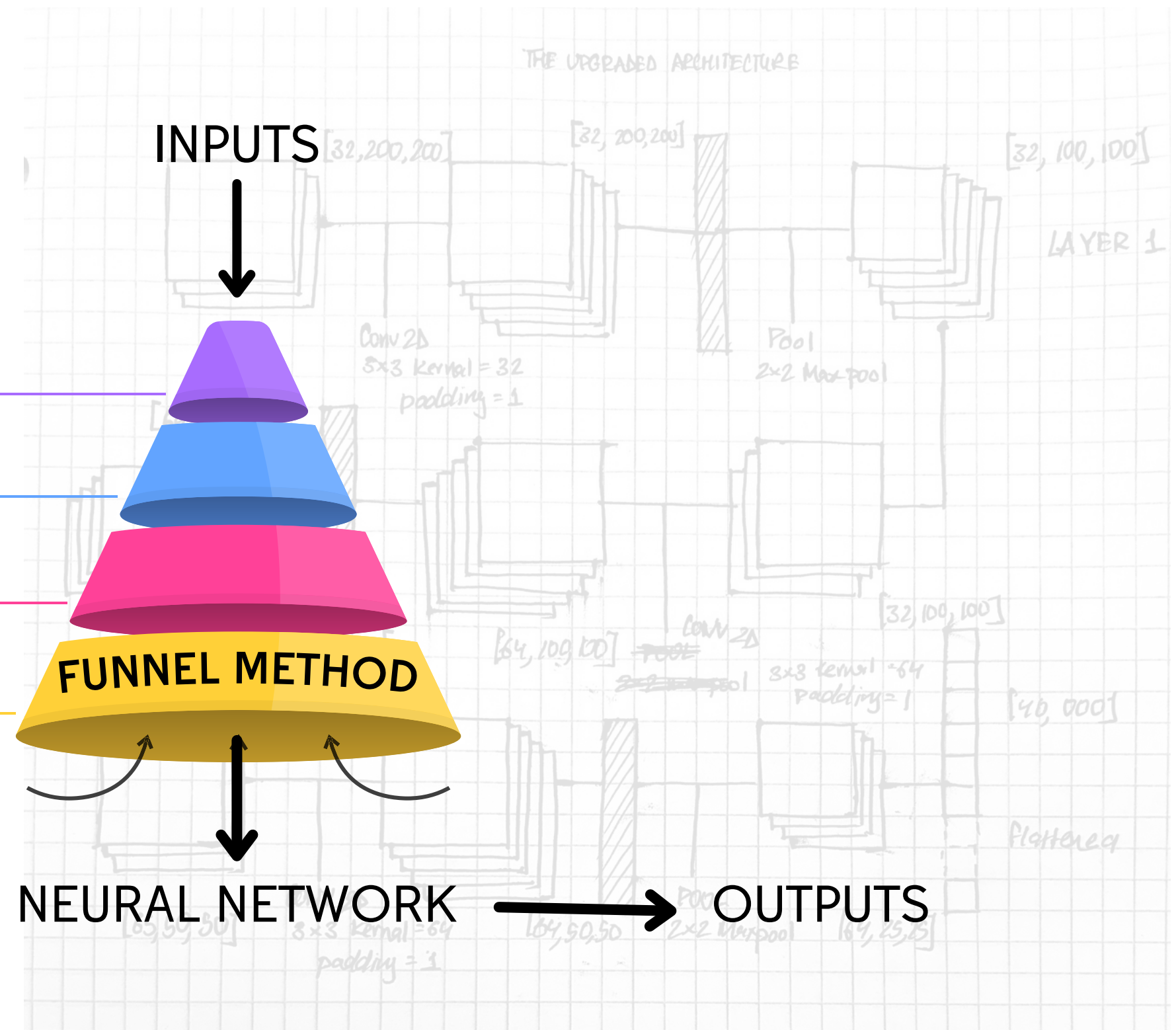
- 1st Layer: Convolution with 32 3x3 kernels
- 2nd Layer: Convolution with 64 3x3 kernels
- 3rd Layer: Convolution with 64 3x3 kernels
- 4th Layer: Convolution with 64 3x3 kernels



New Method

The initial model we designed had the following features :

- 1st Layer: Convolution with 32 3x3 kernels
- 2nd Layer: Convolution with 64 3x3 kernels
- 3rd Layer: Convolution with 64 3x3 kernels
- 4th Layer: Convolution with 64 3x3 kernels



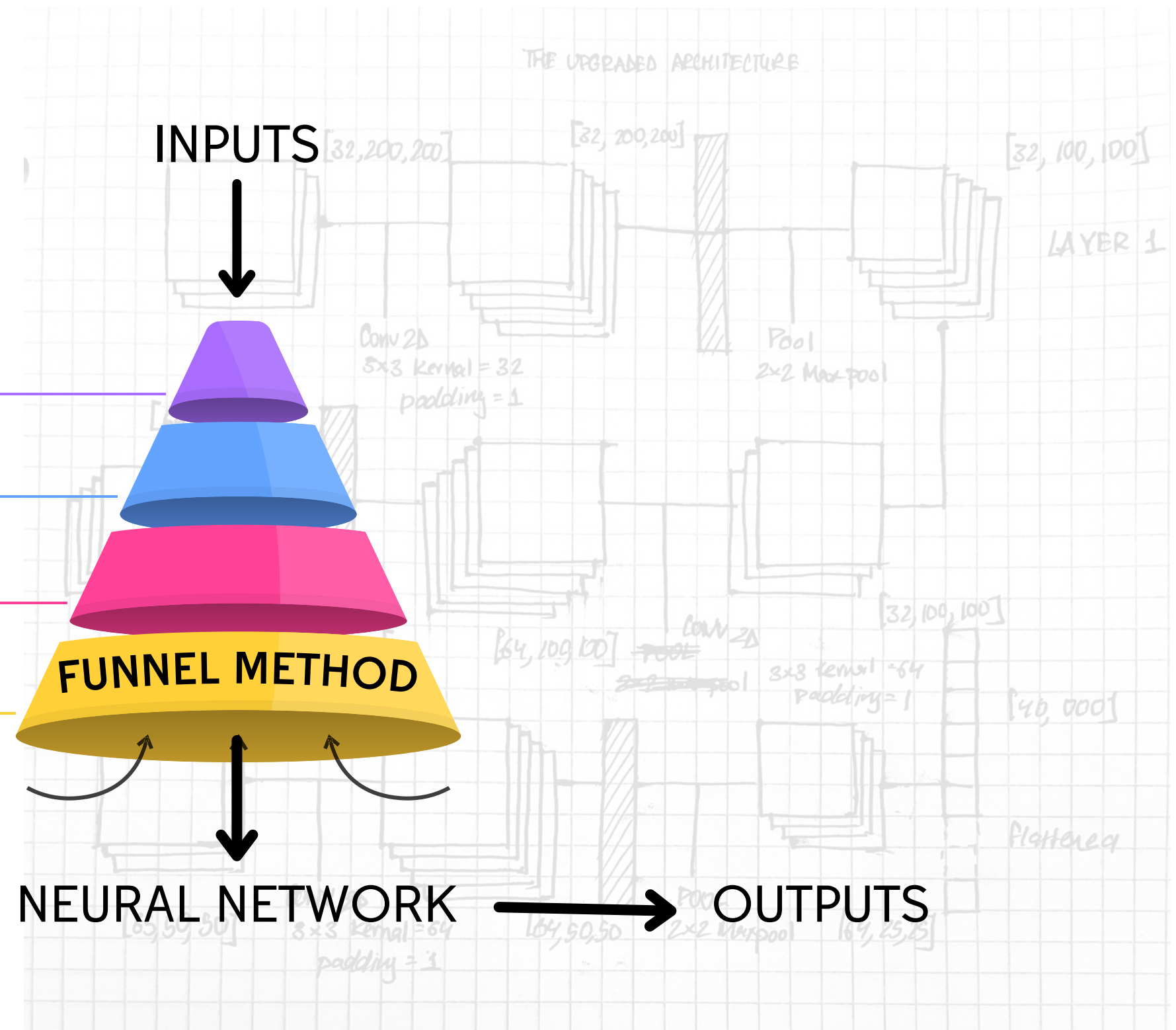
New Method

The initial model we designed had the following features :

- 1st Layer: Convolution with 32 3x3 kernels
- 2nd Layer: Convolution with 64 3x3 kernels
- 3rd Layer: Convolution with 64 3x3 kernels
- ~~4th Layer: Convolution with 64 3x3 kernels~~

Also uses:

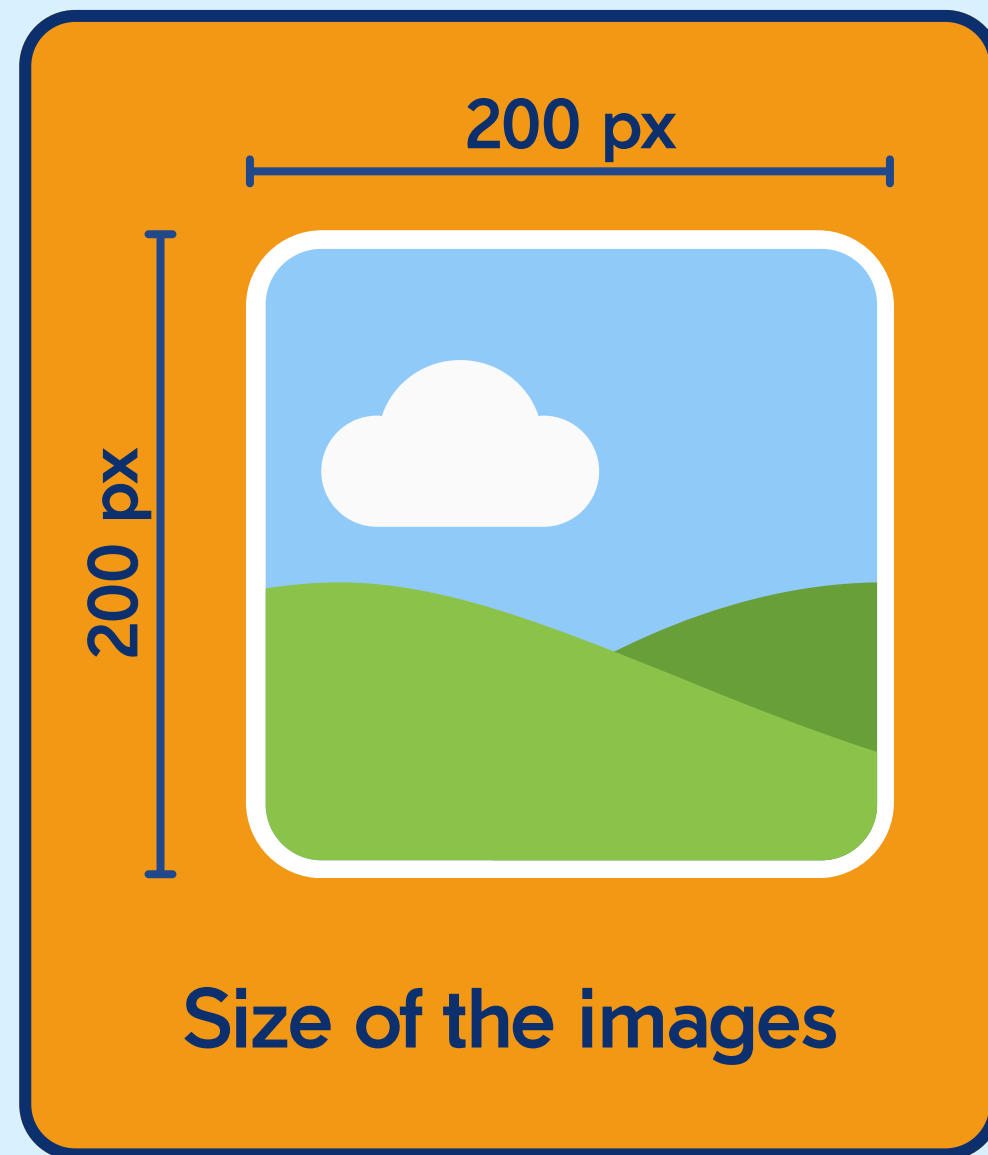
- Optimiser: Adam
- Learning step: 0.001
- Loss Function: Cross Entropy Loss



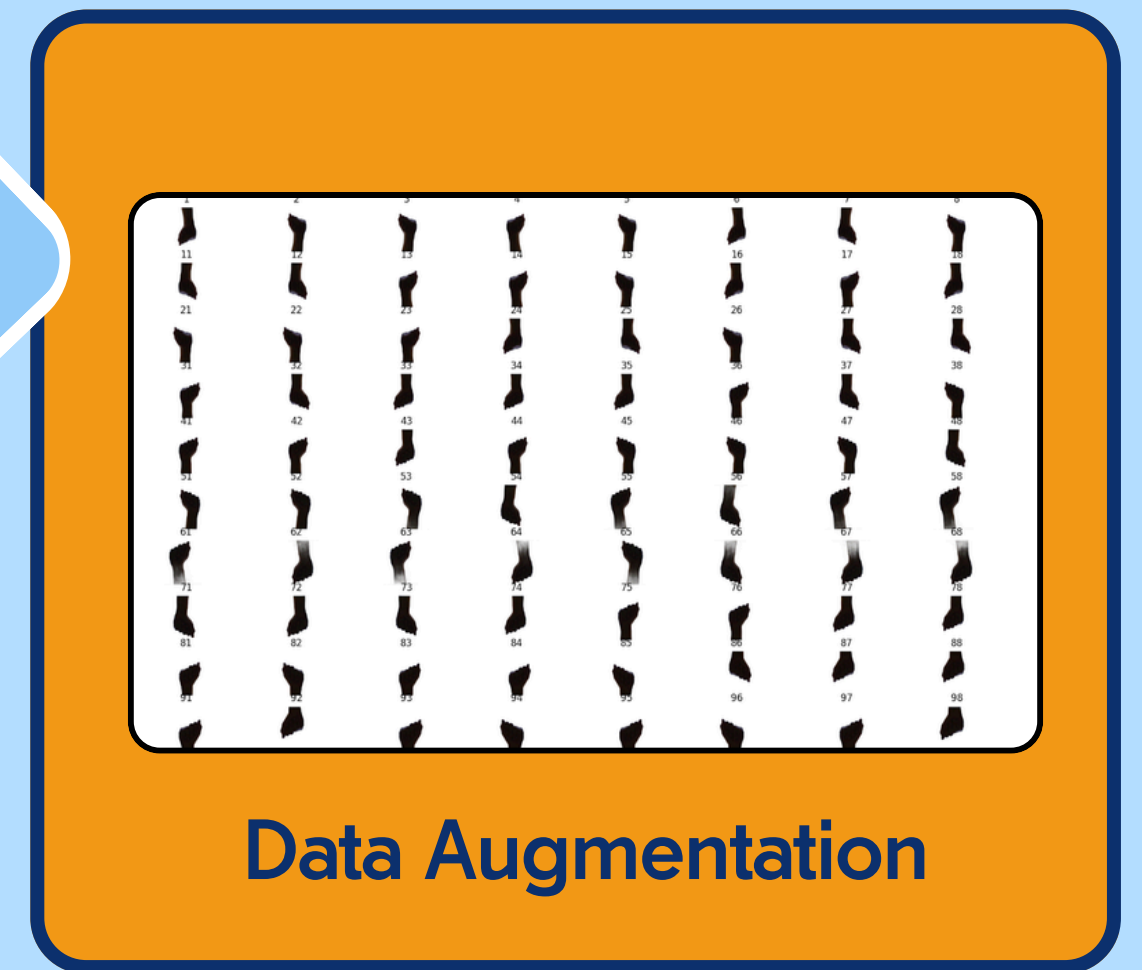
Dataset Processing and Analysis



Characters of Data



Characters of Data

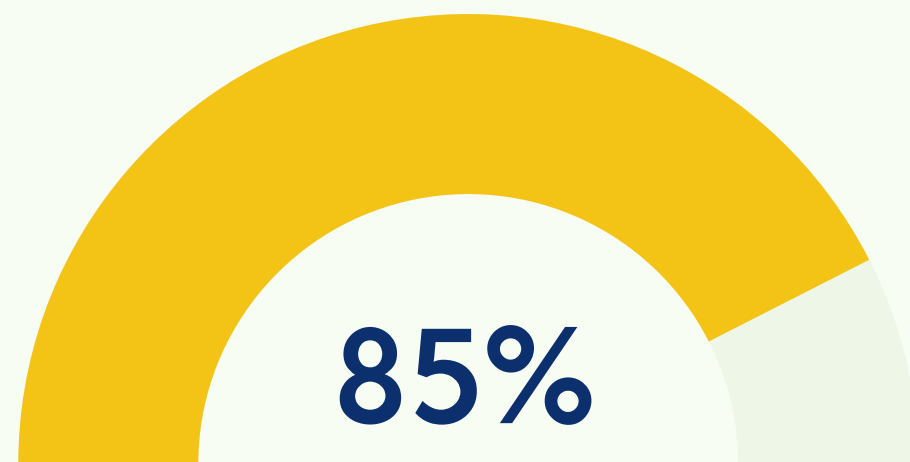




Demo

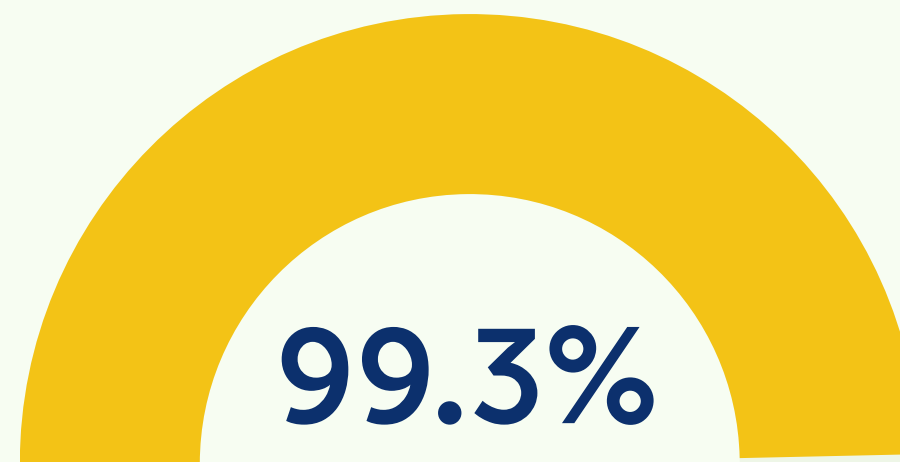
Accuracy

Calculated using Cross Entropy Loss



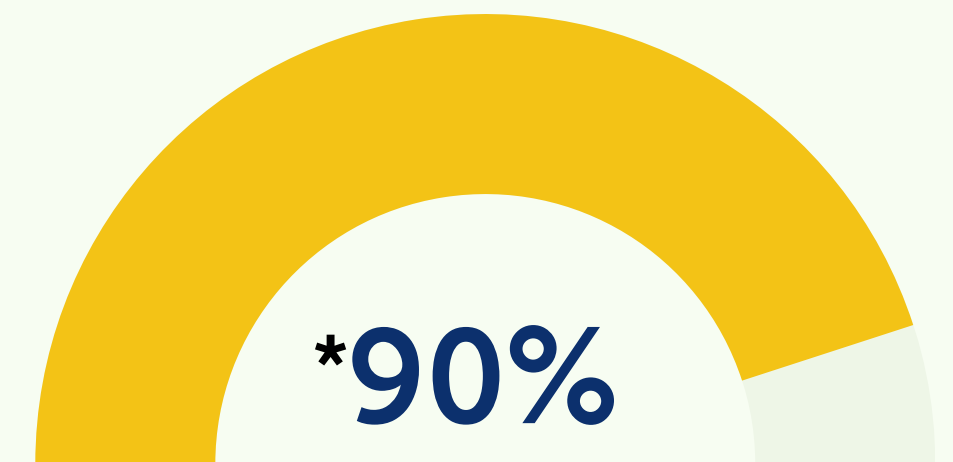
PREDICTION

Estimated using general
data available



INITIAL MODEL

Calculated using
Training Data



FINAL MODEL

Calculated using
Testing Data

Challenges / Reflection

Main Challenges

- ➊ Issues with Overfitting in the Model
- ➋ Applying Proper Data Augmentation
- ➌ Splitting Data Over Training and Testing
- ➍ Model Generation and Implementation

Reflection

Things that we'd do differently:

- ➊ Iterate more with Epochs and CNN Layers
- ➋ Spread out the data throughout the Epochs
- ➌ Add more conventionally used words in ASL

By PySigns

Have a Question?



Thank You.

MARCH 2025

PYSIGNS

[https://github.com/natanielj/
CMPM17-Final-ASLDecoder](https://github.com/natanielj/CMPM17-Final-ASLDecoder)